



Statistics of Financial Markets

Seminar 13: Machine learning



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Acronyms used in this material

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- BET: Bucharest Exchange Trading (index)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- DM: Diebold–Mariano (test)
- EODHD: EOD Historical Data (market-data provider)
- GB: Gradient Boosting
- HAR: Heterogeneous AutoRegressive (model)
- HS: Historical Simulation
- ML: Machine Learning
- MLP: MultiLayer Perceptron
- MSE: Mean Squared Error
- OLS: Ordinary Least Squares
- OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- QGB: Quantile Gradient Boosting
- QLIKE: Quasi-LIKElihood (loss function)
- QR: Quantile Regression
- RF: Random Forest
- ROC: Receiver Operating Characteristic (curve)
- SPX: S&P 500 index (ticker symbol)
- SSE: Sum of Squared Errors
- VIX: CBOE Volatility IndeX

Today's Question and Route

- **Question:** when does a flexible prediction method beat a simple model on financial data, and how do we test it without fooling ourselves?
 - ▶ this seminar comes **before** Lecture 13: the section “What You Need for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: bias and variance, a tree split, ridge and lasso in one dimension, walk-forward and purged designs, on paper
 - ▶ Part B: volatility forecasts, the sign of returns and VaR 1% on real data, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	bias, variance and expected error from five training sets	Solved, Proposed	A1
A3, A4	the best split of a regression tree, step by step	Solved, Proposed	A3
A5, A6	ridge and lasso with one feature	Solved, Proposed	A5
A7, A8	a walk-forward design; a purged K-fold with an embargo	Solved, Proposed	A7
B1, B2	volatility forecasts: S&P 500 (next day), DAX (next week)	Solved, Proposed	B1
B3, B4	the sign of daily returns: DAX; TLV and SNP	Solved, Proposed	B3
B5, B6	VaR 1% of the DAX by quantile regression and quantile boosting	Solved, Proposed	B5
C1, C2	does the S&P 500 help to forecast BET volatility? what is wrong in an AI answer?	Proposed	B2, A7

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Prices used	Period
S&P 500	EODHD	open, high, low, close	2008–2026
DAX	EODHD	open, high, low, close	2006–2026
BET	EODHD	close	2008–2026
Banca Transilvania (TLV), OMV Petrom (SNP)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; daily variance proxy v_t from open, high, low and close ([Chapter 8](#)); the last day is 18 September 2026
- In the notebook: `rv_frame('dax')`, `walk_forward(...)`, `rv_metrics(...)`, `sign_walk_forward('dax')`, `var_backtest(...)` (scikit-learn); no account or key is needed

What You Need for Today (1/5): Learning from Data

- ▣ **Machine learning** (ML): choose a prediction rule $\hat{f}$ from data so that $\hat{f}(x)$ is close to y on **new** data
 - ▶ x_t : the **features** known at the close of day t ; y_t : the **target**; loss: e.g. the squared error $(y - \hat{y})^2$, MSE its mean
- ▣ **Overfitting**: a small error on the training data and a large error on new data
 - ▶ a **hyperparameter** (tree depth, penalty) controls flexibility and is chosen on data not used for the fit
- ▣ **Bias–variance**: if $y = f(x) + \varepsilon$, $\text{Var}(\varepsilon) = \sigma^2$, and $\hat{f}$ is fitted on a random training sample
 - ▶ $E[(y_0 - \hat{f}(x_0))^2] = (E\hat{f}(x_0) - f(x_0))^2 + \text{Var}(\hat{f}(x_0)) + \sigma^2$: bias² + variance + noise
 - ▶ from M training sets: bias = $\bar{f} - f$, variance = $\frac{1}{M} \sum_m (\hat{f}_m - \bar{f})^2$, with $\bar{f}$ the mean forecast

What You Need for Today (2/5): Validation for Time Series

- **Walk-forward:** train on the past, test on the next block (here: one calendar year), refit, move forward
 - ▶ expanding window: the training sample keeps all earlier days
- **Purging:** a target y_t built from days $t + 1, \dots, t + h$ overlaps the test block if t is one of the last h training days: drop them
 - ▶ **embargo:** in K-fold, also drop a few training days right after the test block
- **Look-ahead bias:** a feature uses information not yet known at time t (future prices, full-sample means)
 - ▶ **leakage:** information about test targets reaches training; **random K-fold** (shuffled folds) causes it with overlapping targets

What You Need for Today (3/5): Ridge and Lasso

- ▣ **Ridge:** minimise $\sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j \beta_j^2$; **lasso:** $\dots + \lambda \sum_j |\beta_j|$ ($\lambda \geq 0$: the penalty)
 - ▶ both trade a little bias for a smaller variance; lasso sets some coefficients exactly to zero
- ▣ **One centred feature**, $S_{xx} = \sum_t x_t^2$, $S_{xy} = \sum_t x_t y_t$, no intercept
 - ▶ OLS: $\hat{\beta} = S_{xy}/S_{xx}$; ridge: $\hat{\beta} = S_{xy}/(S_{xx} + \lambda)$
 - ▶ lasso: $\hat{\beta} = \text{sign}(S_{xy}) \max(|S_{xy}| - \lambda/2, 0)/S_{xx}$ (soft thresholding)
- ▣ Standardise the features with the mean and standard deviation of the training sample before the penalty

What You Need for Today (4/5): Trees, Ensembles and Networks

- **Regression tree**: split at $x_j \leq s$; each leaf predicts the mean of its y ; choose (j, s) to minimise
 - ▶ $\text{SSE}(s) = \sum_{x_j \leq s} (y - \bar{y}_L)^2 + \sum_{x_j > s} (y - \bar{y}_R)^2$; candidates: midpoints between sorted values
- **Random forest** (RF): the average of many deep trees, each grown on a bootstrap sample with random subsets of features
 - ▶ **gradient boosting** (GB): small trees added one by one, each fitted to the current residuals, scaled by a learning rate
- **MLP** (multilayer perceptron): $\hat{y} = c + \sum_k v_k g(b_k + w_k^\top x)$ with $g(z) = \max(0, z)$ (ReLU)
 - ▶ **HAR** (heterogeneous autoregressive) benchmark: OLS of $\ln$ future variance on $\ln$ of the variance of the last day, week and month

What You Need for Today (5/5): Evaluation

- Out-of-sample R^2 against a benchmark $\tilde{y}$: $R_{OOS}^2 = 1 - \sum (y_t - \hat{y}_t)^2 / \sum (y_t - \tilde{y}_t)^2$; QLIKE: $L_t = \text{RV}_t / \hat{h}_t + \ln \hat{h}_t$
 - ▶ DM (Diebold–Mariano) test of $d_t = L_t^A - L_t^B$: $\bar{d}/\text{se}(\bar{d}) \sim N(0, 1)$; negative: A is better
- Classification: accuracy against the **majority-class baseline** acc_0 ; $z = (\widehat{\text{acc}} - \text{acc}_0) / \sqrt{\text{acc}_0(1 - \text{acc}_0)/n}$
 - ▶ AUC: the area under the ROC curve ([Chapter 12](#)); $0.5 = \text{no skill}$
- **Quantile regression**: $\hat{q}_\alpha(x) = x^\top \hat{\beta}_\alpha$ minimises $\sum_t \rho_\alpha(y_t - x_t^\top \beta)$, $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
 - ▶ VaR 1% = $-\hat{q}_{0.01}$, a positive loss; backtest with Kupiec (rate) and Christoffersen (independence), [Chapter 10](#)

A1: Bias and Variance of a Shallow Tree [Solved]

Task

- ▣ At a point x_0 the true value is $f(x_0) = 2.0$ and $\sigma^2 = 0.25$. A depth-1 tree fitted on five training sets forecasts 1.6, 1.7, 1.5, 1.8, 1.6.
- ▣ 1. Compute the mean forecast and the bias.
- ▣ 2. Compute the variance of the five forecasts (divide by 5).
- ▣ 3. Compute the expected squared error at x_0 .
- ▣ Report: four numbers and one sentence.

Solution

- ▣ 1. $\bar{f} = 8.2/5 = 1.640$; bias = $1.640 - 2 = -0.360$
- ▣ 2.
 $[(-0.04)^2 + 0.06^2 + (-0.14)^2 + 0.16^2 + (-0.04)^2]/5 = 0.0104$
- ▣ 3. $0.1296 + 0.0104 + 0.25 = 0.390$
- ▣ The shallow tree is stable but systematically too low: its error is mostly squared bias.

A2: Bias and Variance of a Deep Tree [Proposed]

Solution discussed in class

Task

- ▣ Same point and noise as in A1. A depth-10 tree on the same five training sets forecasts 2.6, 1.4, 2.3, 1.7, 2.2. Model: A1.
- ▣ 1. Compute the mean forecast, the bias and the variance.
- ▣ 2. Compute the expected squared error.
- ▣ 3. Decide which tree you would use at x_0 .
- ▣ Report: four numbers and one sentence.

A3: One Split of a Regression Tree [Solved]

Task

- Eight days: x = yesterday's absolute return (%), y = today's variance proxy:
- x : 0.2, 0.4, 0.5, 0.9, 1.1, 1.6, 2.0, 2.8; y : 0.5, 0.7, 0.6, 1.0, 1.4, 2.2, 2.6, 3.4
- 1. Compute the mean of y and the SSE without a split.
- 2. For each candidate split, compute both leaf means and the total SSE.
- 3. Choose the best split and the share of the SSE it removes.
- Report: the table, the split and one sentence.

Solution

s	$\bar{y}_L$	$\bar{y}_R$	SSE
0.30	0.500	1.700	6.740
0.45	0.600	1.867	5.593
0.70	0.600	2.120	3.668
1.00	0.700	2.400	2.220
1.35	0.840	2.733	1.279
1.80	1.067	3.000	2.393
2.40	1.286	3.400	4.089

- $\bar{y} = 1.550$, $SSE = 8.000$; best $s = 1.35$: leaves 0.840 and 2.733, $SSE = 0.532 + 0.747 = 1.279$
- the split removes $1 - 1.279/8.000 = 0.840$ of the SSE: large moves yesterday, high variance today

A4: A Split on the Volatility Index [Proposed]

Solution discussed in class

Task

- ▣ Eight months: x = the VIX level at the end of the month, y = next month's realised volatility (% per day). Model: A3.
- ▣ x : 12, 14, 15, 17, 21, 24, 30, 38; y : 0.8, 0.9, 0.7, 1.1, 1.5, 1.4, 2.6, 3.0
- ▣ 1. Compute the SSE without a split and for every candidate split.
- ▣ 2. Choose the best split and give both leaf forecasts.
- ▣ 3. Compute the share of the SSE that the split removes.
- ▣ Report: the table, three numbers and one sentence.

A5: Ridge with One Feature [Solved]

Task

- ▣ One centred feature with $S_{xx} = 50$ and $S_{xy} = 20$; no intercept.
- ▣ 1. Compute the OLS coefficient.
- ▣ 2. Compute the ridge coefficient and the shrinkage factor $S_{xx}/(S_{xx} + \lambda)$ for $\lambda = 10$ and $\lambda = 50$.
- ▣ 3. Explain what happens when $\lambda \rightarrow \infty$.
- ▣ Report: five numbers and one sentence.

Solution

- ▣ 1. $\hat{\beta}^{OLS} = 20/50 = 0.40$
- ▣ 2. $\lambda = 10$: $20/60 = 0.333$, factor 0.833; $\lambda = 50$: $20/100 = 0.20$, factor 0.50
- ▣ 3. The coefficient tends to 0 but never equals 0.
- ▣ Ridge multiplies the OLS coefficient by a factor below 1: it shrinks, it does not select.

A6: Lasso with One Feature [Proposed]

Solution discussed in class

Task

- ▣ Same data as in A5: $S_{xx} = 50$, $S_{xy} = 20$. Model: A5.
- ▣ 1. Compute the lasso coefficient for $\lambda = 10, 30$ and 50 .
- ▣ 2. Find the smallest λ for which the coefficient is exactly zero.
- ▣ 3. Compare the lasso with the ridge results of A5 for $\lambda = 50$.
- ▣ Report: four numbers and one sentence.

A7: A Walk-Forward Design [Solved]

Task

- ▣ Daily S&P 500 features from 8 April 2008 to 18 September 2026; target: the mean variance of the next 5 days; refits each January from 2013.
- ▣ 1. Count the folds of the walk-forward evaluation.
- ▣ 2. For the 2013 fold, say which training days must be purged and why.
- ▣ 3. Explain why no embargo is needed.
- ▣ 4. Explain what would go wrong if the 2013 model were tuned by random 5-fold on 2008–2012.
- ▣ Report: two numbers and three sentences.

Solution

- ▣ 1. One fold per year 2013–2026: 14 folds.
- ▣ 2. The last 5 training days (from 24 December 2012): their targets use January 2013; the fold trains on 1 188 days, the last one 21 December 2012, and tests on 252.
- ▣ 3. All training days come before the test year: no training feature uses test information.
- ▣ 4. Random folds put neighbouring days with overlapping targets in training and validation: the chosen hyperparameters reward memorising, not forecasting.

A8: A Purged K-Fold with an Embargo [Proposed]

Solution discussed in class

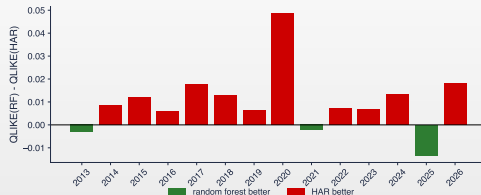
Task

- ▣ 2500 days, $K = 5$ blocks in time order, a target that uses the next 21 days, an embargo of 1% of the days. Model: A7.
- ▣ 1. Compute the size of each test block and of the embargo.
- ▣ 2. Compute the number of training days when the test block is in the middle.
- ▣ 3. Compute it when the test block is the first and when it is the last block.
- ▣ Report: four numbers and one sentence.

B1: Next-Day Volatility of the S&P 500 [Solved]

- **Question:** does a random forest forecast tomorrow's variance of the S&P 500 better than HAR?
- **Inputs:** S&P 500 open, high, low and close since 2008; target: $\ln v_{t+1}$; features: the three HAR averages
- **Tasks**
 1. Fit HAR (OLS) and a random forest (300 trees, at least 20 days per leaf) walk-forward, refitting each January from 2013.
 2. Compute the out-of-sample R^2 of both against the training mean and of the forest against HAR.
 3. Compute the mean QLIKE of both and the Diebold–Mariano test of equal QLIKE.
 4. Draw the QLIKE difference per year.
 5. Interpretation: is the forest worth its complexity here?
- **Report:** four numbers, a test with its p-value, the chart and two sentences
- **Notebook:** section B1

B1: Solution [Solved]



- 3448 days from 2 January 2013; R^2_{OOS} against the mean: HAR 54.8%, forest 53.4%; forest against HAR: -3.2%
- QLIKE: HAR 0.329, forest 0.339; DM = $+2.65$ (p 0.008); the forest is better in only 3 of 14 years
- Interpretation: no: with the same three features the forest is significantly worse than the linear HAR; the relation is close to linear in logs and the forest adds variance

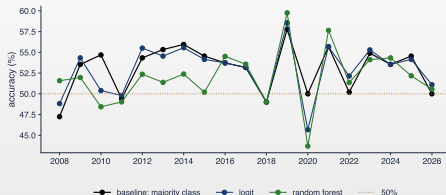
B2: Weekly Volatility of the DAX [Proposed]

- **Question:** do lasso, random forest, gradient boosting or an MLP beat HAR for next week's DAX variance? Model: B1.
- **Inputs:** DAX open, high, low and close since 2006; target: $\ln$ of the mean v over the next 5 days; 11 features (HAR, 4 lags, quarter, r_t , $\min(r_t, 0)$, $|r_t|$)
- **Tasks**
 1. Fit the five models walk-forward from 2012, purging the last 5 training days.
 2. Compute the out-of-sample R^2 against HAR and the mean QLIKE of each model.
 3. Run the Diebold–Mariano test of each model against HAR.
 4. Interpretation: which model would you use for the DAX?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: The Sign of DAX Returns [Solved]

- **Question:** can a logit or a random forest predict whether the DAX rises tomorrow better than “always up”?
- **Inputs:** DAX closes since 2000; target $1\{r_{t+1} > 0\}$; features: the last 5 returns, sums over 5, 21 and 63 days, volatility over 21 and 63 days
- **Tasks**
 1. Fit both classifiers walk-forward from 2008, refitting each January.
 2. Compute the accuracy of each model and of the majority-class baseline.
 3. Test each accuracy against the baseline with the z statistic, and compute the AUC.
 4. Draw the accuracy by year.
 5. Interpretation: does either model have skill?
- **Report:** six numbers, the chart and two sentences
- **Notebook:** section B3

B3: Solution [Solved]



- $n = 4752$ days; up days 53.1% = the baseline accuracy; logit 52.9% ($z = -0.20$, $p = 0.839$, AUC 0.507); forest 52.0% ($z = -1.45$, $p = 0.146$, AUC 0.508)
- The logit forecasts “up” on 90% of the days; the forest beats the baseline in 8 of 19 years, as often as chance would allow
- Interpretation: no: both accuracies are at or below the baseline, and the AUC is close to 0.5; daily DAX returns are close to unpredictable (Chapter 7)

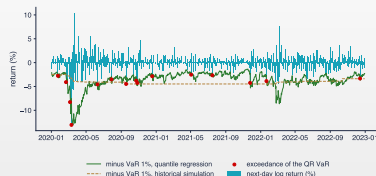
B4: The Sign of Two BVB Stocks [Proposed]

- **Question:** is the sign of Banca Transilvania and OMV Petrom returns easier to predict than that of the DAX? Model: B3.
- **Inputs:** TLV (without 30–31 May 2016) and SNP adjusted closes since 2010; the features of B3; forecasts from 2014
- **Tasks**
 1. Fit the logit, random forest and gradient boosting walk-forward from 2014.
 2. Compute the accuracy, the baseline accuracy, the z statistic and the AUC for each stock and model.
 3. Interpretation: is there more predictability on the BVB than on the DAX?
- **Report:** one table and two sentences
- **Notebook:** section B4

B5: VaR 1% of the DAX by Quantile Regression [Solved]

- **Question:** does a linear quantile regression on volatility features give a better VaR 1% than historical simulation?
- **Inputs:** DAX since 2006; target r_{t+1} ; features: $\sqrt{v_t}$, $\sqrt{\text{mean } v}$ over 5 and 22 days, r_t
- **Tasks**
 1. Fit the 1% quantile regression walk-forward from 2012 and set $\text{VaR } 1\% = -\hat{q}_{0.01}$.
 2. Compute the historical-simulation VaR 1% on the last 500 days.
 3. Count the exceptions and run the Kupiec and Christoffersen tests; compute the average quantile loss.
 4. Draw both VaR forecasts in 2020–2022.
 5. Interpretation: which VaR would you report to a risk committee?
- **Report:** a table, the chart and two sentences
- **Notebook:** section B5

B5: Solution [Solved]



Method	x	rate (%)	p Kupiec	p ind.	loss $\times 100$
QR	39	1.05	0.783	0.364	3.80
HS	49	1.31	0.066	< 0.001	4.75

- 3 731 days from 2 January 2012; last fit: $\hat{q}_{0.01} = -0.472 - 0.434\sqrt{v_t} - 1.814\sqrt{v^{(w)}} - 0.221\sqrt{v^{(m)}} + 0.158r_t$; VaR 1% on 17 September 2026: 2.24%
- Interpretation: the quantile regression: right rate, independent exceptions, smaller loss; HS is exceeded in clusters (13 times in 2020) because a 500-day window reacts slowly

B6: Quantile Gradient Boosting for the DAX [Proposed]

- **Question:** does a flexible quantile model improve on the linear quantile regression of B5? Model: B5.
- **Inputs:** as in B5; gradient boosting with the quantile loss at 1%, 300 trees of depth 2, learning rate 0.05, at least 50 days per leaf
- **Tasks**
 1. Fit the quantile boosting walk-forward from 2012 and compute its VaR 1%.
 2. Count the exceptions, run the Kupiec and Christoffersen tests and compute the quantile loss.
 3. Interpretation: is the flexible model worth it at the 1% level?
- **Report:** one table and two sentences
- **Notebook:** section B6

C1: Does the S&P 500 Help to Forecast BET Volatility? [Proposed]

- **Question:** the BVB closes before New York; does yesterday's S&P 500 volatility improve a HAR forecast of next week's BET variance?
- **Inputs:** BET and S&P 500 closes since 2008; proxy: squared daily returns (the BET has no high and low prices in our data); models: B1, B2
- **Tasks**
 1. Build the HAR features of the BET and the HAR features of the S&P 500 at the previous New York close.
 2. Compare HAR, HAR plus the S&P 500 features (OLS) and a random forest on all features, walk-forward from 2012.
 3. Compute the out-of-sample R^2 , QLIKE and the Diebold–Mariano tests against HAR.
 4. Interpretation: is the S&P 500 information worth adding?
- **Report:** a table and a plan for a project
- **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- ▣ A student asked an AI assistant to summarise machine learning for the S&P 500. The answer:
- ▣ (a) With random 5-fold cross-validation a random forest explains 36.9% of the next 21-day return: the market is predictable.
- ▣ (b) The forest fits next week's log variance better than HAR: in-sample R^2 80.2% against 73.7%.
- ▣ (c) A forest predicts the sign of tomorrow's return correctly on 53.1% of the days, more than a coin: it has skill.
- ▣ (d) For the VaR 99% use quantile regression at $\tau = 0.99$ on the next-day return.
- ▣ (e) The lasso removes the features whose t-statistics are not significant.
- ▣ (f) QLIKE punishes an under-forecast of the variance more than an over-forecast by the same factor.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ The expected error is $\text{bias}^2 + \text{variance} + \text{noise}$: flexibility is not free
- ▣ Time series: walk-forward with purging; never shuffle days with overlapping targets
- ▣ Volatility: linear and penalised models are hard to beat; trees on the HAR features alone lose
- ▣ The sign of daily returns: compare with “always up”; no model beats it here
- ▣ VaR 1%: a linear quantile regression passes the backtests; a flexible quantile model at 1% does not
- ▣ An AI answer is a draft: check the validation scheme, the baseline and the VaR convention

After the Seminar

- ▣ Lecture 13 develops each topic of today: bias and variance, validation, ridge and lasso, trees and ensembles, neural networks, the three applications and data snooping
- ▣ Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- ▣ C1 can grow into a team project: BVB stocks, external markets, several horizons
- ▣ Reading: [James et al. \(2023\)](#), Ch. 2, 5, 6 and 8; [Franke, Härdle & Hafner \(2019\)](#), Ch. 19

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